Chapter 9. Fourier Series

Orthonormal systems, pointwise convergence, Fejér's theorem, Parseval's identity, smoothness and decay

Contents

9.1 Orthonormal systems **2**

9.2 Pointwise convergence **2**

9.3 Fejér's theorem and Parseval's identity **3**

9.4 Smoothness, decay, differentiation and integration **4**

Throughout, functions are 2π-periodic and complex valued, $\mathcal{R}^1 = \mathcal{R}^1[-\pi,\pi]$, and

$$\hat f(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt, \qquad s_N(f;x) = \sum_{n=-N}^{N} \hat f(n)e^{inx}.$$

The **Fourier series** of f is $\sum_{n\in\mathbb{Z}} \hat f(n)e^{inx}$; in real form $\frac{a_0}{2} + \sum(a_n\cos nx + b_n \sin nx)$ with $a_n = \hat f(n) + \hat f(-n)$, $b_n = i(\hat f(n) - \hat f(-n))$.

9.1 Orthonormal systems

In a complex inner product space X, $\{e_n\}$ is **orthonormal** if $\langle e_m, e_n\rangle = \delta_{mn}$. With $\langle f,g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f\bar g$, the functions $e_n(x) = e^{inx}$ form an orthonormal system in $\mathcal{R}^2$, and $\hat f(n) = \langle f, e_n\rangle$.

Theorem 9.1 (Best approximation and Bessel's inequality)

For $v \in X$ and an orthonormal system $\{e_n\}$, put $c_n = \langle v, e_n\rangle$. Among all finite combinations $\sum_{|n|\le N} a_n e_n$, the distance to v is minimized exactly by $a_n = c_n$, and

$$\Big\| v - \sum_{|n|\le N} c_n e_n \Big\|^2 = \|v\|^2 - \sum_{|n|\le N} |c_n|^2.$$

Consequently $\sum_n |c_n|^2 \le \|v\|^2$ (Bessel). In particular $\sum |\hat f(n)|^2 \le \frac{1}{2\pi}\int |f|^2$ for $f \in \mathcal{R}^2$.

Proof. Expanding, $\|v - \sum a_n e_n\|^2 = \|v\|^2 - \sum|c_n|^2 + \sum|a_n - c_n|^2$. ∎

So $s_N(f)$ is the best L^2 approximation of f by trigonometric polynomials of degree N. Basic rules: $\widehat{f(\cdot - a)}(n) = e^{-ina}\hat f(n)$; $\widehat{f'}(n) = in\hat f(n)$ for C^1 periodic f (integration by parts); and the convolution $f * g(x) = \frac{1}{2\pi}\int f(x-t)g(t)dt$ satisfies $\widehat{f*g} = \hat f \hat g$.

Example 9.1

(a) $f(x) = x$ on $(-\pi,\pi)$: $\hat f(0) = 0$, $\hat f(n) = \frac{(-1)^n i}{n}$, so $f \sim 2\sum_{n\ge1}(-1)^{n+1}\frac{\sin nx}{n}$.
(b) $f(x) = |x|$ on $[-\pi,\pi]$: $f \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k\ge0}\frac{\cos(2k+1)x}{(2k+1)^2}$, converging uniformly by the M-test.
At $x = 0$: $\sum \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$, hence $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$.

9.2 Pointwise convergence

Theorem 9.2 (Riemann–Lebesgue lemma)

If $\int_I |f| < \infty$ on an interval I (possibly infinite), then $\int_I f(t)e^{i\lambda t}dt \to 0$ as $|\lambda| \to \infty$. In particular $\hat f(n) \to 0$.

Proof. For the indicator of an interval $[\alpha,\beta]$ the integral is $\frac{e^{i\lambda\beta} - e^{i\lambda\alpha}}{i\lambda} \to 0$, hence the claim holds for step functions; for general f approximate in $\|\cdot\|_1$ by step functions (Theorem 7.9). ∎

The partial sums are convolutions with the **Dirichlet kernel**:

$$s_N(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_N(t)dt, \qquad D_N(t) = \sum_{|n|\le N} e^{int} = \frac{\sin(N+\frac12)t}{\sin\frac{t}{2}}, \qquad \frac{1}{2\pi}\int D_N = 1.$$

Theorem 9.3 (Localization)

If $f \in \mathcal{R}^1$ and $f = 0$ on an interval around x, then $s_N(f;x) \to 0$. Hence the convergence of $s_N(f;x)$ and its limit depend only on the values of f near x.

Proof. If $f(x-t) = 0$ for $|t| < \delta$, then $s_N(f;x) = \frac{1}{2\pi}\int_{\delta \le |t| \le \pi} \frac{f(x-t)}{\sin(t/2)} \sin(N+\frac{1}{2})t\,dt$, and $f(x-t)/\sin(t/2)$ is integrable there; apply Riemann–Lebesgue. ∎

Theorem 9.4 (Dini's test)

Let $f \in \mathcal{R}^1$ and $s \in \mathbb{C}$. If $\int_0^\delta \frac{|f(x+t)+f(x-t)-2s|}{t} dt < \infty$ for some $\delta > 0$, then $s_N(f;x) \to s$.

Proof. Using the evenness of D_N, $s_N(f;x) - s = \frac{1}{2\pi}\int_0^\pi \{f(x+t)+f(x-t)-2s\} D_N(t) dt$, and $\frac{f(x+t)+f(x-t)-2s}{\sin(t/2)}$ is integrable on $(0,\pi]$ by hypothesis (since $\sin(t/2) \ge t/\pi$). Riemann–Lebesgue finishes the proof. ∎

Corollary 9.5

If f satisfies a Lipschitz (or Hölder) condition at x, or is differentiable at x, then $s_N(f;x) \to f(x)$. If f has one-sided limits $f(x\pm)$ and one-sided derivatives at a jump, then $s_N(f;x) \to \frac{1}{2}\{f(x+)+f(x-)\}$.

Theorem 9.6 (Dirichlet–Jordan)

If f is of bounded variation on an interval around x, then $s_N(f;x) \to \frac{1}{2}\{f(x+)+f(x-)\}$; if f is continuous and BV on a closed interval, the convergence is uniform on compact subintervals of its interior.

Sketch. By localization and the Jordan decomposition one may assume $g(t) = f(x+t)+f(x-t)-f(x+)-f(x-)$ is increasing on $[0,\delta]$ with $g(0+) = 0$. By the second mean value theorem for integrals (Chapter 5), $\int_0^\eta g(t) D_N(t) dt = g(\eta-)\int_c^\eta D_N$ for some c, and $|\int_c^\eta D_N|$ is bounded independently of N, c, η (because $\int_0^y \frac{\sin u}{u} du$ is bounded). Choosing η with $g(\eta-)$ small and applying Riemann–Lebesgue on $[\eta,\pi]$ gives the result. ∎

Example 9.2

(a) For $f(x) = x$ on $(-\pi,\pi)$, Dini's test gives $x = 2\sum(-1)^{n+1}\frac{\sin nx}{n}$ on $(-\pi,\pi)$, and the series is $0 = \frac{1}{2}(f(\pi-)+f(-\pi+))$ at $\pm\pi$. Setting $x = \frac{\pi}{2}$ gives Leibniz's series for $\frac{\pi}{4}$.
(b) Continuity alone does not guarantee convergence: there are continuous functions whose Fourier series diverge at a point (du Bois-Reymond), essentially because $\frac{1}{2\pi}\int|D_N| \sim \frac{4}{\pi^2}\log N \to \infty$.
(c) Near a jump, s_N overshoots by about 9% of the jump no matter how large N is (Gibbs phenomenon), so the convergence in Corollary 9.5 is not uniform near jumps.

9.3 Fejér's theorem and Parseval's identity

The Cesàro means $\sigma_N = \frac{1}{N+1}\sum_{k=0}^N s_k$ are convolutions with the **Fejér kernel**

$$K_N(t) = \frac{1}{N+1}\sum_{k=0}^{N} D_k(t) = \frac{1}{N+1}\Big(\frac{\sin\frac{N+1}{2}t}{\sin\frac{t}{2}}\Big)^2,$$

which, unlike D_N, is nonnegative, has $\frac{1}{2\pi}\int K_N = 1$, and satisfies $\sup_{\delta\le|t|\le\pi} K_N(t) \to 0$ for every $\delta > 0$ (an approximate identity).

Theorem 9.7 (Fejér)

If $f \in \mathcal{R}^1$ has one-sided limits at x, then $\sigma_N(f;x) \to \frac{1}{2}\{f(x+) + f(x-)\}$. If f is continuous (and periodic), $\sigma_N(f) \to f$ uniformly.

Proof. $\sigma_N(f;x) - s = \frac{1}{2\pi}\int_0^\pi \{f(x+t) + f(x-t) - 2s\}K_N(t)dt$. Split at δ: on $[0,\delta]$ the bracket is small and $K_N \ge 0$ has integral at most 1; on $[\delta,\pi]$, $K_N \to 0$ uniformly and f is integrable. For continuous f, uniform continuity makes δ independent of x. ∎

Corollary 9.8

(a) Trigonometric polynomials are uniformly dense in the continuous periodic functions. (b) If the Fourier series of a continuous periodic f converges at x, its sum is $f(x)$. (c) If $\hat f(n) = 0$ for all n and f is continuous, then $f = 0$.

Proof. (a) $\sigma_N(f)$ is a trigonometric polynomial. (b) If $s_N \to s$, then $\sigma_N \to s$ (Cesàro), while $\sigma_N \to f(x)$. (c) $\sigma_N(f) = 0$. ∎

Theorem 9.9 (Parseval's identity)

For $f, g \in \mathcal{R}^2$: $\|f - s_N(f)\|_2 \to 0$,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f|^2 = \sum_{n\in\mathbb{Z}} |\hat f(n)|^2, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} f\overline{g} = \sum_n \hat f(n)\overline{\hat g(n)}.$$

Proof. Given ϵ, choose continuous periodic h with $\|f - h\|_2 < \epsilon$ (Theorem 7.9) and a trigonometric polynomial P of degree M with $\|h - P\|_\infty < \epsilon$ (Corollary 9.8). For $N \ge M$, by best approximation, $\|f - s_N(f)\|_2 \le \|f - P\|_2 < 2\epsilon$. The identity follows from Theorem 9.1, and the polarized form from the polarization identity. ∎

Example 9.3

Parseval for $f(x) = x$: $\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{3} = \sum_{n\ne 0}\frac{1}{n^2}$, again $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$. For $f(x) = x^2$ one obtains $\sum \frac{1}{n^4} = \frac{\pi^4}{90}$.

9.4 Smoothness, decay, differentiation and integration

Theorem 9.10 (Term-by-term integration)

For $f \in \mathcal{R}^2$ and any a, x, $\int_a^x f = \sum_n \hat f(n)\int_a^x e^{int}dt$, the series converging uniformly in x; this holds even if the Fourier series of f itself diverges.

Proof. $\left|\int_a^x (f - s_N(f))\right| \le \sqrt{2\pi}\cdot\sqrt{2\pi}\|f - s_N(f)\|_2 \to 0$ by Cauchy–Schwarz and Parseval. ∎

Smoothness of f corresponds to decay of $\hat f$. If $f \in C^k$ (periodic) then $\widehat{f^{(k)}}(n) = (in)^k \hat f(n)$, so $|\hat f(n)| = o(|n|^{-k})$ by Riemann–Lebesgue. If f is C^1, then $\sum|\hat f(n)| \le |\hat f(0)| + \left(\sum \frac{1}{n^2}\right)^{1/2}\left(\sum |n\hat f(n)|^2\right)^{1/2} < \infty$ and the Fourier series converges absolutely and uniformly to f.

Theorem 9.11

A periodic f is C^∞ iff $\hat f$ is rapidly decreasing ($|n|^k \hat f(n) \to 0$ for every k). Conversely every rapidly decreasing $a : \mathbb{Z} \to \mathbb{C}$ is $\hat f$ for the C^∞ function $f(x) = \sum a(n) e^{inx}$. A periodic f is real analytic iff $\limsup |\hat f(n)|^{1/|n|} < 1$.

Proof. The first statements follow from the paragraph above and Theorem 6.4 (the differentiated series converge uniformly). For analytic f, f extends holomorphically to a strip $|\operatorname{Im} z| < r$, and shifting the contour in $\hat f(n)$ gives $|\hat f(n)| \le C e^{-r'|n|}$ for $r' < r$; conversely geometric decay makes $\sum \hat f(n) e^{inz}$ converge in a strip. ∎

Not every null sequence is a Fourier coefficient sequence of an $\mathcal{R}^1$ (or even L^1) function: if f is odd and integrable, $\sum_{n \ge 1} \frac{\hat f(n)}{n}$ converges (it equals, up to a constant, $\int_0^\pi$ of the integral of f), whereas for $b_n = 1/\log n$ the series $\sum \frac{b_n}{n}$ diverges. So $\sum \frac{\sin nx}{\log n}$, which converges everywhere by Theorem 6.13, is not the Fourier series of an integrable function. Chapter 10 completes the picture with L^1, L^2 and the Riesz–Fischer theorem.